

Potential Fields

Robots in Context

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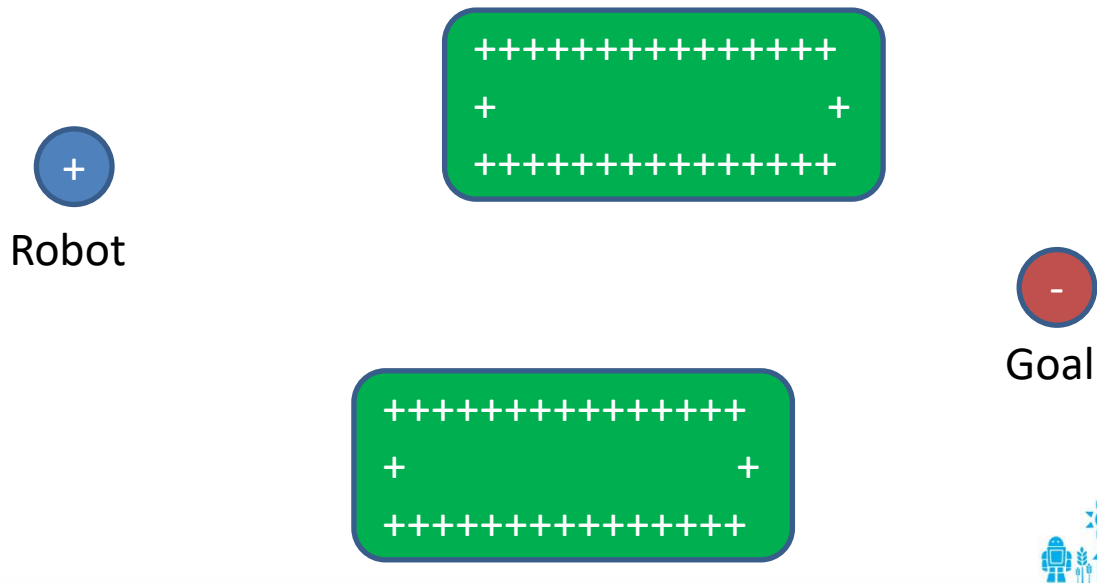
Content

- Potential functions
 - Attractive and Repulsive potentials
 - Gradient descent
 - Brushfire
- The Wavefront planner
- Last part:
 - Roadmaps



Potential Functions

- Basic idea
 - The goal is attracting the robot
 - Objects are repelling the robot



“Simple” Potential Functions

- Attractive Potential
 - $U_{att}(q) = \frac{1}{2} \gamma d(q, q_{goal})^2$
 - Where $d(q, q_{goal})$ is a distance function and γ determines the strength of the attractive potential



"Simple" Potential Functions

- Attractive Potential
 - $U_{att}(q) = \frac{1}{2} \gamma d(q, q_{goal})^2$
 - Where $d(q, q_{goal})$ is a distance function and γ determines the strength of the attractive potential
- Repulsive Potential
 - $U_{rep}(q) = \begin{cases} \frac{1}{2} \mu (\frac{1}{D(q)} - \frac{1}{Q^*})^2, & D(q) \leq Q^* \\ 0, & D(q) > Q^* \end{cases}$
 - Where $D(q)$ is the distance to the nearest obstacle, μ determines the strength of the repulsive potential and Q^* is a distance threshold.



Exercise: Draw the potentials assuming q is 2-dimensional

- Attractive Potential

- $U_{att}(q) = \frac{1}{2} \gamma d(q, q_{goal})^2$

- Where $d(q, q_{goal})$ is a distance function and γ determines the strength of the attractive potential

- Repulsive Potential

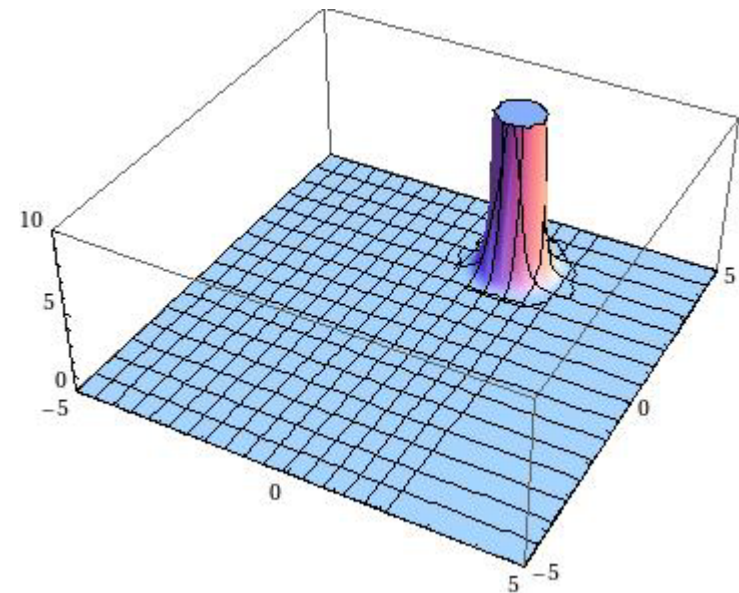
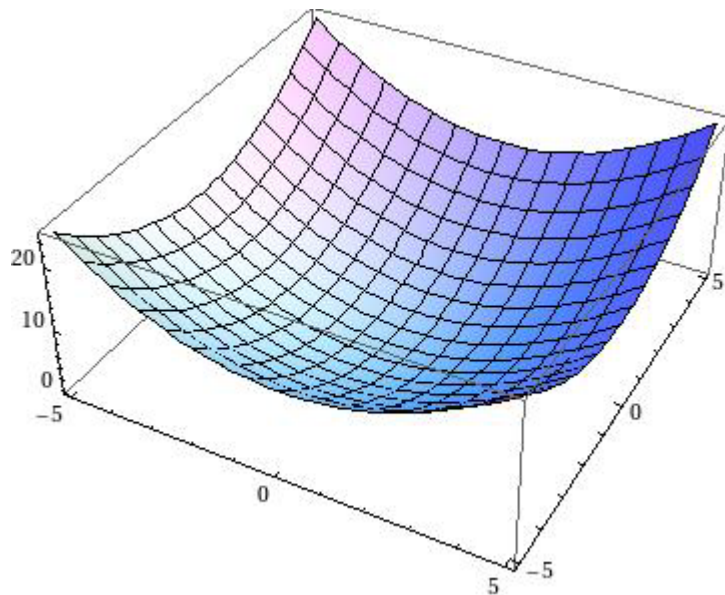
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Exercise: Draw the potentials assuming
q is 2-dimensional

- Attractive Potential
- Repulsive



Choosing reasonable parameters, the previous expressions can be visualized directly



Combining attractive and repulsive potentials

- Add the two potentials
 - $U(q) = U_{att}(q) + U_{rep}(q)$
 - In case of multiple obstacles we can either use
 - $U(q) = U_{att}(q) + \sum_{i=1}^N U_{rep,i}(q)$
- or
- $U(q) = U_{att}(q) + \min_i U_{rep}(q)$



- Combined attractive and repulsive potential





-



Gradients

What is the gradient of the attractive potential

$$U_{att}(q) = \frac{1}{2} \gamma \, d(q, q_{goal})^2$$

What is the gradient of the repulsive potential

$$U_{rep}(q) = \begin{cases} \frac{1}{2}\mu \left(\frac{1}{D(q)} - \frac{1}{Q^*} \right)^2, & D(q) \leq Q^* \\ 0, & D(q) > Q^* \end{cases}$$



Gradients

What is the gradient of the attractive potential $U_{att}(q) = \frac{1}{2} \gamma d(q, q_{goal})^2$

$$\nabla U_{att}(q) = \frac{1}{2} \gamma \nabla d(q, q_{goal})^2 = \gamma (q - q_{goal})$$



Gradients

What is the gradient of the attractive potential $U_{att}(q) = \frac{1}{2} \gamma d(q, q_{goal})^2$

$$\nabla U_{att}(q) = \frac{1}{2} \gamma \nabla d(q, q_{goal})^2 = \gamma (q - q_{goal})$$

What is the gradient of the repulsive potential $U_{rep}(q) =$

$$\begin{cases} \frac{1}{2} \mu \left(\frac{1}{D(q)} - \frac{1}{Q^*} \right)^2, & D(q) \leq Q^* \\ 0, & D(q) > Q^* \end{cases}$$

$$\nabla U_{rep}(q) = \begin{cases} \mu(\frac{1}{Q^*} - \frac{1}{D(q)}) \frac{1}{D(q)^2} \nabla D(q), & D(q) \leq Q^* \\ 0, & D(q) > Q^* \end{cases}$$



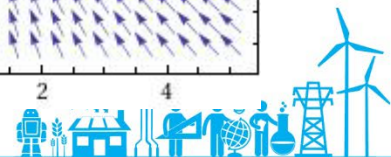
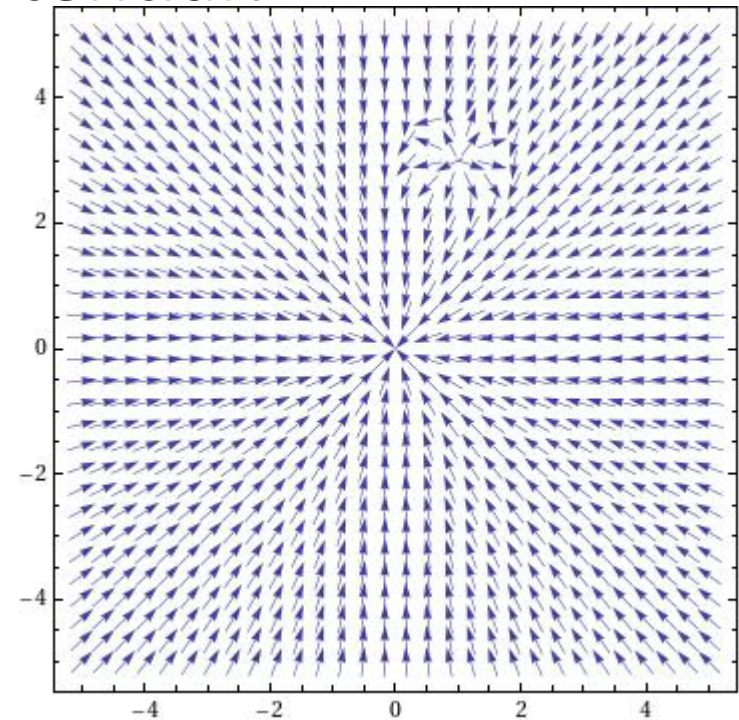
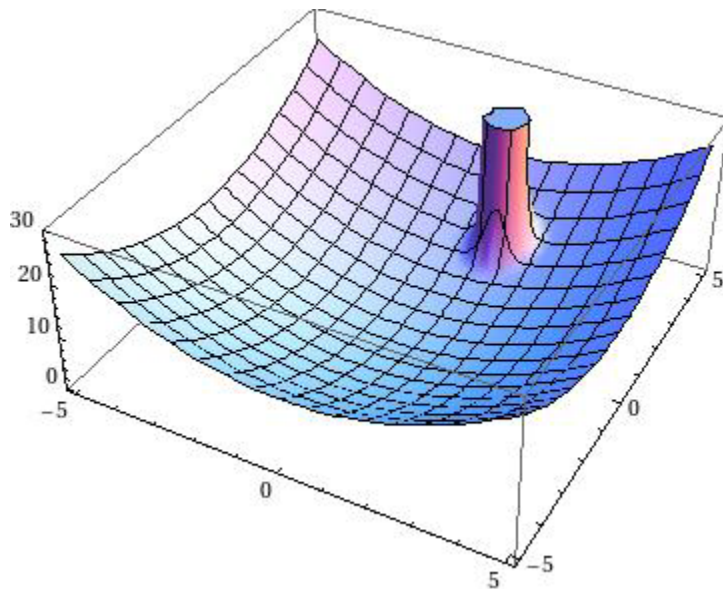
Gradients

- What is the physical interpretation of a first order derivative of a potential?



Gradient Combined

- What is the physical interpretation of a first order derivative of a potential?



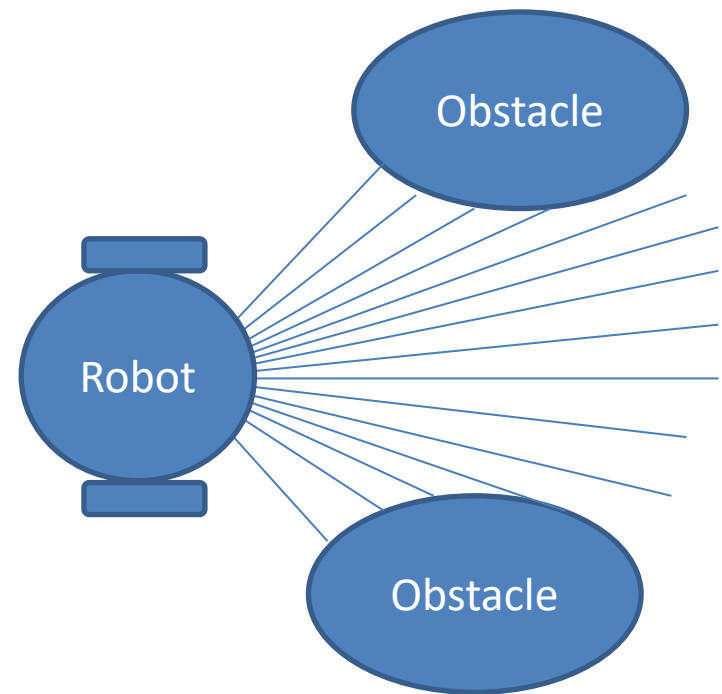
Algorithm Characteristics

- Completeness properties?
 - Any particular scenarios where it and where it might get stuck?
- Online vs. Offline
- Sensor based vs. Model based
- Greedy vs. exhaustive



Sensor Based Potential Field

- Use e.g. laser scanner to detect obstacles.
- Use the closest reading as the obstacle in the potential.
- The gradient of the repulsive potential is given based on the direction of the reading.



Model Based Potential Field

The Brushfire Algorithm

- Assume that we have a regular grid with cells either being free (value 0) or obstacles (value 1).

			1						
			1						
			1		1	1	1		
							1	1	
							1	1	



Model Based Potential Field

The Brushfire Algorithm

- We now wish to define the distance function $D(q)$ for each cell

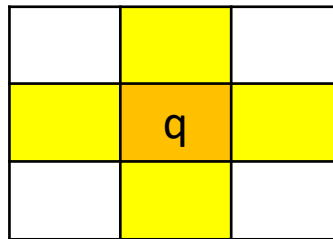
			1						
			1						
			1		1	1	1		
							1	1	
							1	1	



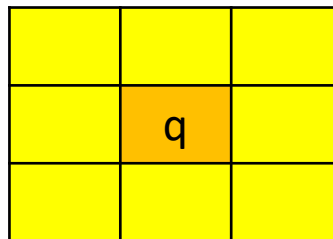
Model Based Potential Field

The Brushfire Algorithm

- Defining the neighbors of a cell
 - 4 Connected: Connect only direct neighbors



- 8 Connected: Connect also diagonally



Model Based Potential Field

The Brushfire Algorithm

- Start labeling all nodes next to an obstacle 2

[illegible]

Model Based Potential Field

The Brushfire Algorithm

- Next label all neighbors to 2's as 3

	3	2	2	2	3				
	3	2	1	2	3	3	3	3	3
	3	2	1	2	2	2	2	2	3
	3	2	1	2	1	1	1	2	3
	3	2	2	2	2	2	2	2	2
	3	3	3	3	3	2	1	1	2
					3	2	1	1	2
					3	2	2	2	2



- And so forth...

2



Model Based Potential Field

The Brushfire Algorithm

- And so forth...
- What about the border region?

4	3	2	2	2	3	4	4	4	4
4	3	2	1	2	3	3	3	3	3
4	3	2	1	2	2	2	2	2	3
4	3	2	1	2	1	1	1	2	3
4	3	2	2	2	2	2	2	2	2
4	3	3	3	3	3	2	1	1	2
4	4	4	4	4	3	2	1	1	2
5	5	5	5	4	3	2	2	2	2



Model Based Potential Field

The Brushfire Algorithm

- The distance to an obstacle can now be defined based on the value of a cell.
- The gradient as the direction to a neighbor with a smaller value.

4	3	2	2	2	3	4	4	4	4
4	3	2	1	2	3	3	3	3	3
4	3	2	1	2	2	2	2	2	3
4	3	2	1	2	1	1	1	2	3
4	3	2	2	2	2	2	2	2	2
4	3	3	3	3	3	2	1	1	2
4	4	4	4	4	3	2	1	1	2
5	5	5	5	4	3	2	2	2	2



The Wave-front planner

- Apply a brush fire like algorithm from the goal.
- Assume that obstacles have value 1 and free space has value 0.
- Algorithm:
 - Assign the goal cell the value 2
 - Add the neighbors of the goal cell to a set L
 - While L is not empty
 - Take an element of L
 - Assign the element the value of its neighbor+1
 - Add all neighbors with 0 value to L



Wavefront planner: setup

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
4	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	
3	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



1

1. *Chlorophyll a* (Chl *a*) and *Chlorophyll b* (Chl *b*) were determined using the method of Lichtenthaler and Wherry (1987). The total chlorophyll content was determined using the method of Lichtenthaler and Wherry (1987). The total chlorophyll content was determined using the method of Lichtenthaler and Wherry (1987).

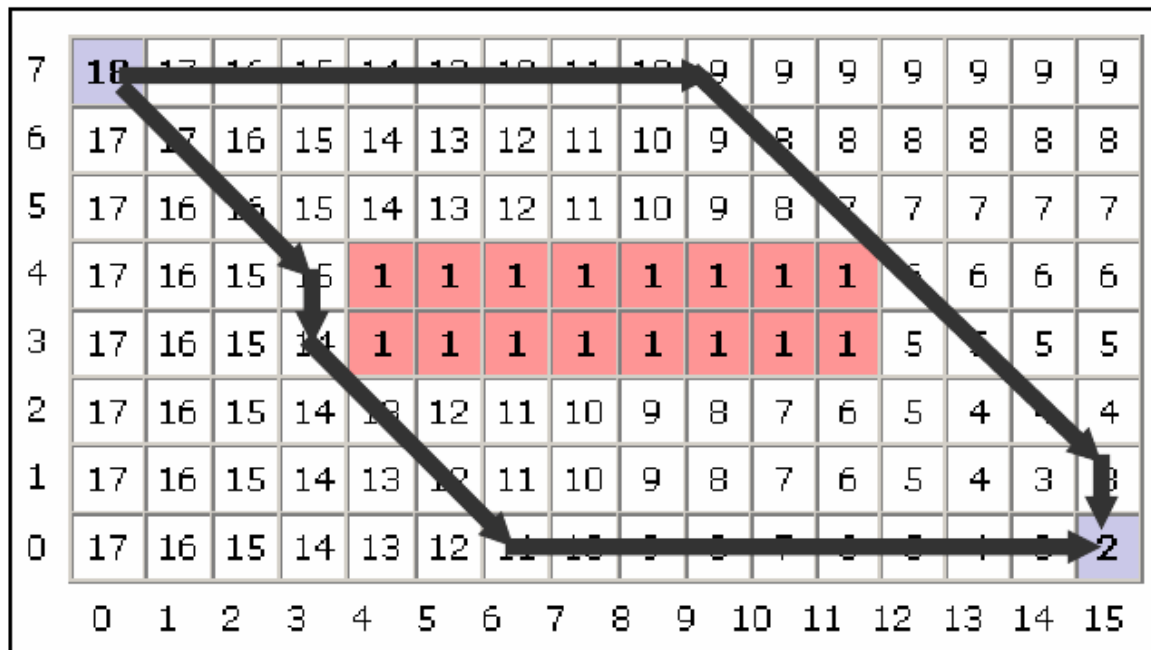


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Wavefront, now what?

- To find the shortest path, according to your metric, simply always move toward a cell with a lower number
 - The numbers generated by the Wavefront planner are roughly proportional to their distance from the goal



Two possible
Shortest paths



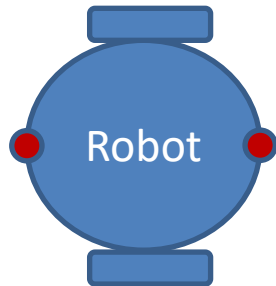
Wavefront Properties

- What can be said about the properties of the algorithm?
 - Completeness?
 - Greedy or exhaustive?



Potential function on rigid body

- Select N points (●) on the robot
 - N should be large enough to control the robot AND represent its shape relative to obstacles.

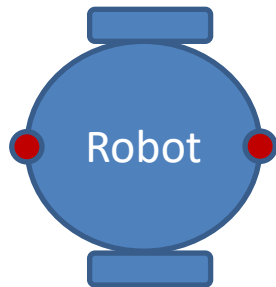


Enough to pin
down the robot

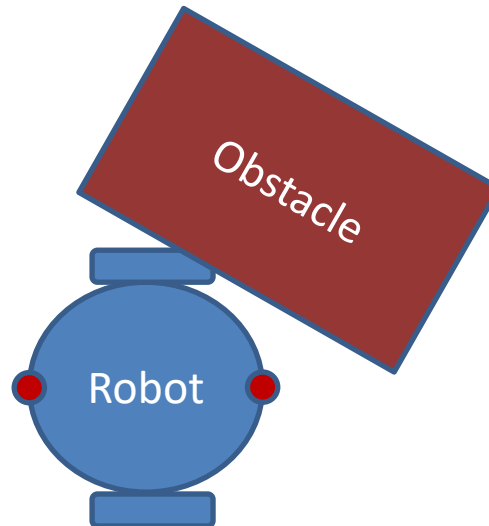


Potential function on rigid body

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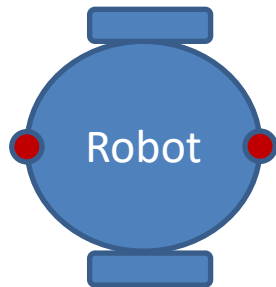


Not enough to avoid collision

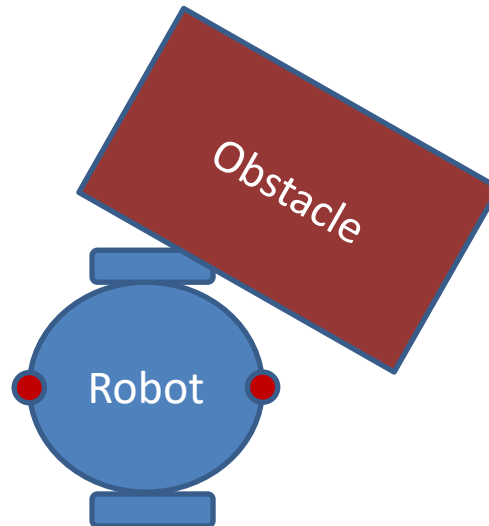


Potential function on rigid body

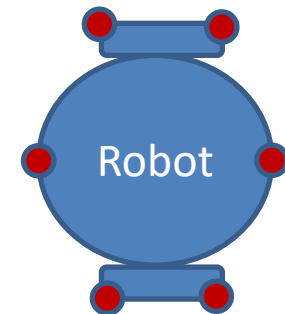
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Enough to pin down the robot



Not enough to avoid collision



When obstacles are "large" point representing the convex hull will be enough

Potential function on rigid body

- Attractive potential for each point

$$U_{\text{att},j}(q) = \begin{cases} \frac{1}{2}\zeta_i d^2(r_j(q), r_j(q_{\text{goal}})), & d(r_j(q), r_j(q_{\text{goal}})) \leq d_{\text{goal}}^* \\ d_{\text{goal}}^* \zeta_j d(r_j(q), r_j(q_{\text{goal}})) - \frac{1}{2}\zeta_j d_{\text{goal}}^*, & d(r_j(q), r_j(q_{\text{goal}})) > d_{\text{goal}}^*. \end{cases}$$

$$\nabla U_{\text{att},j}(q) = \begin{cases} \zeta_i (r_j(q) - r_j(q_{\text{goal}})), & d(r_j(q), r_j(q_{\text{goal}})) \leq d_{\text{goal}}^*, \\ \frac{d_{\text{goal}}^* \zeta_j (r_j(q) - r_j(q_{\text{goal}}))}{d(r_j(q), r_j(q_{\text{goal}}))}, & d(r_j(q), r_j(q_{\text{goal}})) > d_{\text{goal}}^*. \end{cases}$$

- Repelling potential for each point

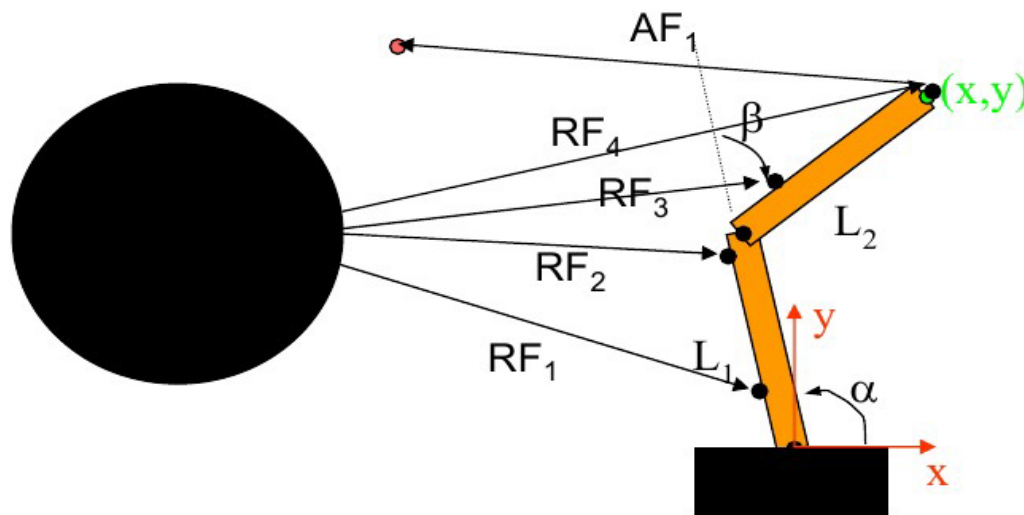
$$U_{\text{repi},j}(q) = \begin{cases} \frac{1}{2}\eta_j \left(\frac{1}{d_i(r_j(q))} - \frac{1}{Q_i^*} \right)^2, & d_i(r_j(q)) \leq Q_i^*, \\ 0, & d_i(r_j(q)) > Q_i^*, \end{cases}$$

$$\nabla U_{\text{repi},j}(q) = \begin{cases} \eta_j \left(\frac{1}{Q_i^*} - \frac{1}{d_i(r_j(q))} \right) \frac{1}{d_i^2(r_j(q))} \nabla d_i(r_j(q)), & d_i(r_j(q)) \leq Q_i^*, \\ 0, & d_i(r_j(q)) > Q_i^*. \end{cases}$$



Potential Fields for Manipulator

- Pick several points on the manipulator
- Compute attractive and repulsive potentials for each
- Transform these into the configuration space
- Use the resulting force to move the robot (in its configuration space)



We continue at:

